

# Persuading an inattentive and privately informed receiver

Pietro Dall'Ara  
*University of Naples Federico II & CSEF*

Grass workshop

# Inattention and persuasion

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This paper studies the persuasion of a **receiver** who is privately informed about costs and benefits of information.

# Overview

This paper studies persuasion à la Kamenica and Gentzkow when Receiver is **privately informed** and **rationally inattentive**.

Do screening devices help Sender? Which types of Receiver pay attention?

Persuasion leverages the **complementarity** between information quality and attention effort.

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Results:

1. Persuasion mechanisms are equivalent to experiments;
2. Upper censorships are optimal in applications;
3. Persuasion mechanisms are without loss under *deterministic* coordination mechanisms,  
but with loss under *stochastic* coordination mechanisms.

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Plan:

1. Model;
2. Literature;
3. Equivalence between persuasion mechanisms and experiments;
4. Optimal experiment with single-peaked outside option;
5. Generalized principal-agent setting (Myerson, 1982).

**Model**

# Model

Sender and Receiver.

Payoff to Receiver from **action**  $a \in \{0, 1\}$  and **effort**  $e \in [0, 1]$  is

$$U_R(a, \theta, e, c, \lambda) = \underbrace{a(\theta - c)}_{\text{material payoff } u_R} - \underbrace{\lambda k(e)}_{\text{effort cost}},$$

in which:

$\theta \in [0, 1]$  is the state, drawn from a continuous  $F_0$ , with mean  $x_0$ ;

$c \in [0, 1]$  is the outside option;

$\lambda \geq 0$  is the attention cost;

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Payoff to Sender from action  $a$  is  $U_S(a) = a$ .

Receiver is privately informed about her **type**:  $(c, \lambda)$ , independent of  $\theta$ , drawn from  $H$  with support  $T$ .

# Persuasion mechanisms

A signal is  $\sigma: \Theta \rightarrow \Delta M$ .

A persuasion mechanism is a list of signals  $(\sigma_r)_{r \in T}$ .

## Timing

1. Sender commits to a persuasion mechanism  $(\sigma_r)_{r \in T}$ ;
2. Receiver chooses a signal  $\sigma_r$  and effort  $e$ , knowing her type;
3. Nature draws  $\theta$  according to  $F_0$ ;  
With probability  $e$ , Receiver observes a message  $m$  drawn from  $\sigma_r(\theta)$ ;
4. Receiver chooses an action  $a$ .

# Literature

## Bayesian persuasion

Kamenica and Gentzkow (2011)... **Kolotilin et al. (2017)**, Kolotilin (2018), Guo and Shmaya (2019), Kolotilin et al. (2022).

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## Persuasion of an inattentive receiver

**Bloedel and Segal (2021), Wei (2021).**

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## **Persuasion with “outside information”**

Bizzotto et al. (2020); Dworczak and Pavan (2022); Matysková and Montes (2023).

## **Dynamic information provision**

Liao (2021); Che et al. (2023); Au and Whitmeyer (2023); Knoepfle (2024); Hébert and Zhong (2025); Jain and Whitmeyer (2025)...

## **Attention management**

Lipnowski et al. (2020).

**Equivalence between persuasion  
mechanisms and signals**

# Mean distribution

Fix a signal.

**Optimal action** given belief  $\mu$  of type- $(c, \lambda)$  Receiver is:

$$a^*(\mu) \in [\mathbf{1}\{\mathbb{E}_{\theta \sim \mu}[\theta] > c\}, \mathbf{1}\{\mathbb{E}_{\theta \sim \mu}[\theta] \geq c\}].$$

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**Marginal benefit of effort** given belief distribution  $p$  of type- $(c, \lambda)$

Receiver is:

$$B(c) = \underbrace{\mathbb{E}_{\mu \sim p}[\mathbb{E}_{\theta \sim \mu}[u_R(a^*(\mu), c, \theta)]]}_{\text{payoff of informed action}} - \underbrace{u_R(a^*(F_0), c, x_0)}_{\text{payoff of uninformed action}}.$$

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The mean distribution describes equilibrium action and effort.

# Signals as information policies

We identify a signal with its **information policy**.

The *information policy* of the CDF  $F$  is the function  $I_F: \mathbb{R}_+ \rightarrow \mathbb{R}_+$  given by

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(1)  $I_F$  is convex;

(2)  $I_{\bar{F}} \leq I_F \leq I_{F_0}$ , in which:

$\bar{F}$  is the CDF induced by an **uninformative signal**.

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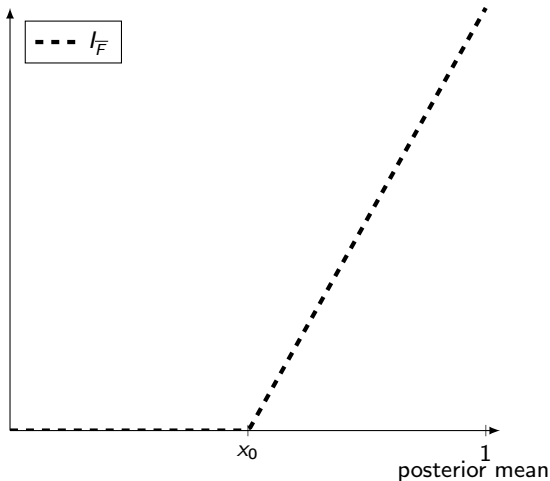
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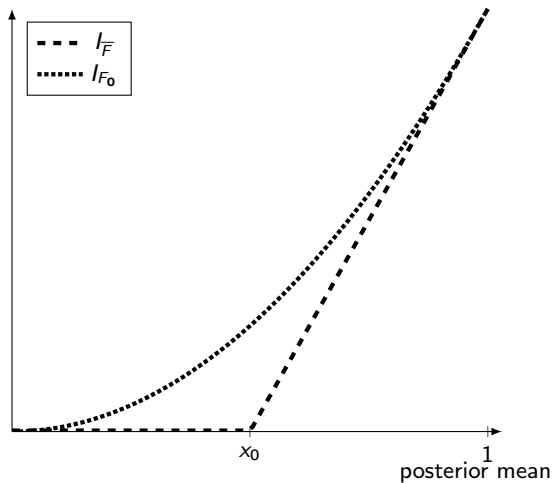
**Lemma 1 (Gentzkow and Kamenica, 2016; Kolotilin, 2018.)**

If  $I: \mathbb{R}_+ \rightarrow \mathbb{R}_+$  satisfies (1) and (2), then there exists a signal that induces a posterior-mean CDF with information policy  $I$ .

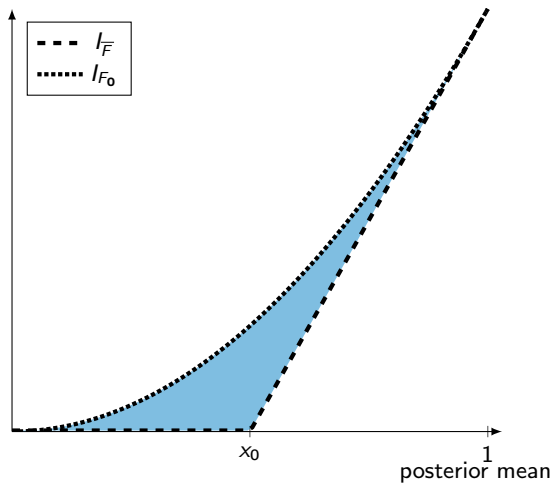
# Information policies



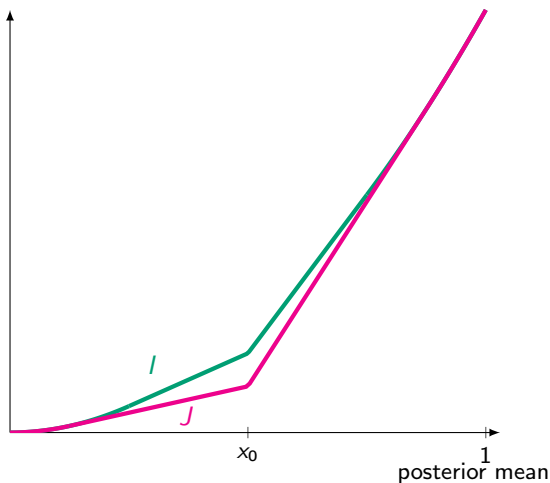
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# Information policies



$I$  is "more informative" than  $J$  if:  $I \geq J$ .

# Interval structure of the extensive margin

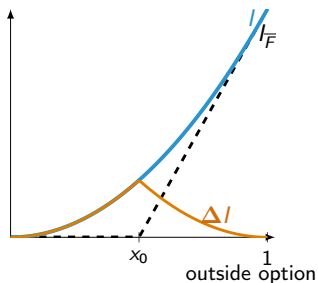
The *net informativeness* of  $I$  is denoted by

$$\Delta I = I - I_{\bar{F}}.$$

## Lemma 2

The marginal benefit of effort given information policy  $I$  satisfies, for all  $c$ ,

$$B(c) = \Delta I(c).$$



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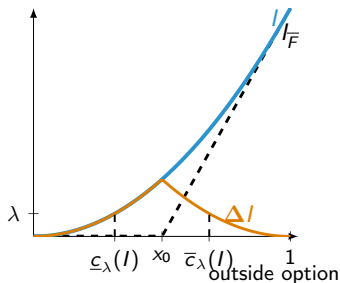
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If **cost is linear** ( $k(e) = e$ ),  
then there are *cutoff types*  
 $c_{\lambda}(I)$ ,  $\bar{c}_{\lambda}(I)$  such that  
intermediate outside options are  
attentive.

# Persuasion mechanisms

The *value of information policy*  $I$  is the interim payoff of type- $(c, \lambda)$  Receiver:

$$V_R(\Delta I(c), \lambda) := \max_{e \in [0,1]} \underbrace{e \Delta I(c)}_{\substack{\text{strictly} \\ \text{increasing} \\ \text{differences}}} - \lambda k(e).$$

- Information and attention are **complements**.

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A *persuasion mechanism* is a menu of information policies:  $(I_r)_{r \in T}$ .

A persuasion mechanism  $(I_r)_{r \in T}$  is *incentive compatible* (IC) if:  
for all types  $(c, \lambda) \in T$  and reports  $\widetilde{(c, \lambda)} \in T$ , we have

$$V_R(\Delta I_{(c, \lambda)}(c), \lambda) \geq V_R(\Delta I_{\widetilde{(c, \lambda)}}(c), \lambda).$$

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► Information and attention are **complements**.

An IC persuasion mechanism  $(I_r)_{r \in T}$  is *equivalent to information policy*  $J$  if, for every type  $(c, \lambda)$ , the optimal effort and action are the same:

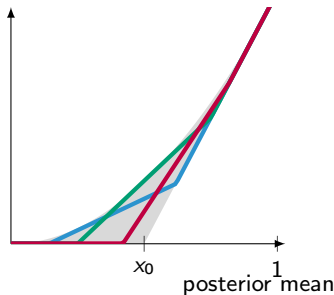
- $\operatorname{argmax}_{e \in [0,1]} e \Delta I_{(c,\lambda)}(c) - \lambda k(e) \subseteq \operatorname{argmax}_{e \in [0,1]} e \Delta J(c) - \lambda k(e),$
- if  $(0, 1] \cap \operatorname{argmax}_{e \in [0,1]} e \Delta I_{(c,\lambda)}(c) - \lambda k(e) \neq \emptyset$ , then  $\partial I_{(c,\lambda)}(c) \subseteq \underbrace{\partial J(c)}_{\substack{[J'_-(c), J'_+(c)] \\ \text{subdifferential}}}.$

# Equivalence

## Proposition 1

For every IC persuasion mechanism, there exists an equivalent information policy.

## Proof sketch



1. By ET, the rent from report  $r$  is  $\int_0^{\Delta I_r(c)} \operatorname{argmax}_e \{eb - \lambda k(e)\} db$ .

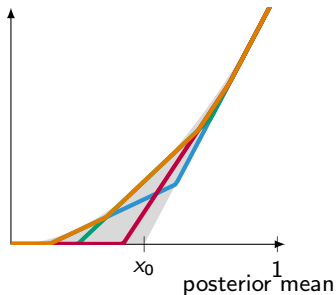
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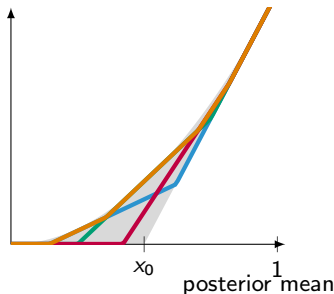
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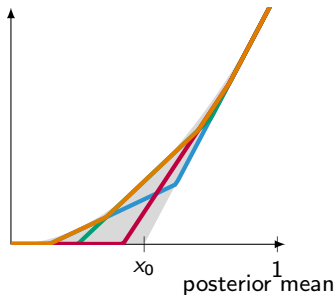
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If  $\lambda = 0$ , then: Kolotilin et al. (2017).

**Optimal information policy**

# Sender's problem

The Sender's expected payoff from information policy  $I$  is

$$V_S(I) = \int_T \underbrace{e^*(I, c, \lambda)}_{\text{extensive margin}} \underbrace{(1 - I'_-(c) - \mathbf{1}\{x_0 \geq c\})}_{\text{intensive margin}} + \mathbf{1}\{x_0 \geq c\} dH(c, \lambda).$$

Assumption: (1) str. convex  $k$ ; (2) Receiver chooses action 1 if indifferent.

$I$  is optimal if  $I$  solves

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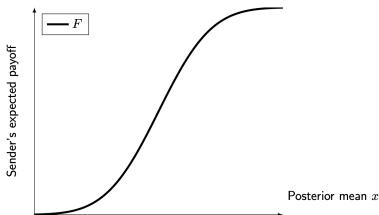
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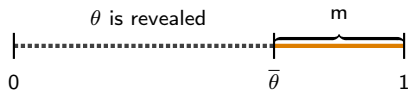
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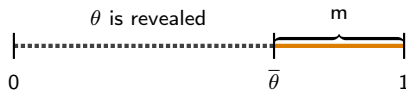
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# Optimal signal



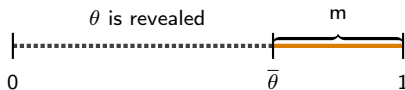
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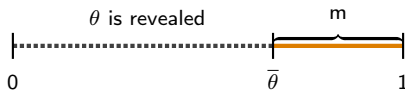
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- By increasing differences, the **effective** information policy  $x \mapsto e^* \cdot I(x) + (1 - e^*) \cdot I_{\bar{F}}(x)$  is nondecreasing in  $I$ ;

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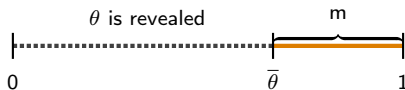
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⌈ Not a linear function of  $I$  (Augias and Uhe, 2025).

# Application to media censorship

- (1) Sender knows the attention cost  $\lambda$ .
- (2)  $k$  is linear.
- (3) Sender's payoff is:

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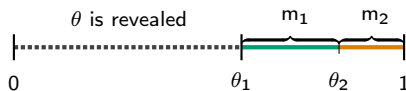
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## Proposition 3

Under (1), (2), (3), and SPness, there exists a bi-upper censorship signal that induces an optimal information policy.



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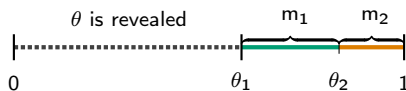
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Kolotilin et al. (2022): no media market;

Gehlbach and Sonin (2014): Sender knows  $c$ , and does not know  $\lambda$ .

**Generalized principal-agent problem and  
coordination mechanisms**

# Generalized principal-agent problem

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- ▶ A coordination mechanism  $(R, M, \pi)$  is *direct* if:  $R = T$ ,  $M = [0, 1]$ ;
- ▶ A direct coordination mechanism  $\pi$  is IC if:
  1. truthful reporting;
  2. obeying the effort recommendation;are optimal for Receiver.

## Proposition 4 (Myerson 1982.)

For every coordination mechanism  $G$ , for every equilibrium Sender's payoff  $v$  in  $G$ , there exists a direct and IC coordination mechanism  $\pi$  such that

$$\underbrace{V(\pi)}_{\text{Sender's payoff in the truthful \& obedient eq'm of } \pi} = v.$$

# Deterministic coordination mechanisms

- ▶ A coordination mechanism  $(R, M, \pi)$  is *deterministic* if there exists  $(I_r)_{r \in T}$  such that, for all  $r$ ,  $\pi(\{I_r\} \times M | r) = 1$ .
- ▶ An IC persuasion mechanism  $(I_r)_{r \in T}$  induces a Sender's payoff  $v$  if:

$$v = \int_T e^*(I_{(c, \lambda)}, c, \lambda)(1 - (I_{(c, \lambda)})'_-(c) - \mathbf{1}\{x_0 \geq c\}) + \mathbf{1}\{x_0 \geq c\} dH(c, \lambda).$$

## Proposition 5

For every deterministic coordination mechanism  $G$ , for every equilibrium Sender's payoff  $v$  in  $G$ , there exists a persuasion mechanism that induces the Sender's payoff  $v$ .

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- ▶ A coordination mechanism  $(R, M, \pi)$  is *deterministic* if there exists  $(I_r)_{r \in T}$  such that, for all  $r$ ,  $\pi(\{I_r\} \times M | r) = 1$ .
- ▶ An IC persuasion mechanism  $(I_r)_{r \in T}$  induces a Sender's payoff  $v$  if:

$$v = \int_T e^*(I_{(c,\lambda)}, c, \lambda)(1 - (I_{(c,\lambda)})'_-(c) - \mathbf{1}\{x_0 \geq c\}) + \mathbf{1}\{x_0 \geq c\} dH(c, \lambda).$$

## Proposition 5

For every deterministic coordination mechanism  $G$ , for every equilibrium Sender's payoff  $v$  in  $G$ , there exists a persuasion mechanism that induces the Sender's payoff  $v$ .

Deterministic coordination mechanisms are equivalent to signals.

# Correlating recommendation and information

**Example:** Linear  $k$ , uniform  $\theta$ ,  $c = 3/4$  known to Sender, and two types:

**Low** ( $\lambda_1 \approx 0.01$ ), Prob =  $8/9$ ;

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There exists a **coordination mechanism** such that

**Low** gets a deterministic recommendation, 1, and some (low) information  $I_L$ .

**High** gets a **lottery**:

- ▶ Prob  $p \approx 0.3$ : Recommend  $e = 1$  and the very informative  $J_H$ ;
- ▶ Prob  $1 - p \approx 0.7$ : Recommend  $e = 0$  and no information;

which improves upon any deterministic coordination mechanism.

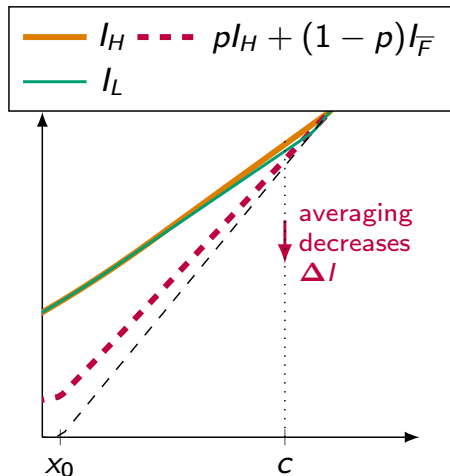
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- ▶ High (weakly) prefers the lottery because  $\Delta I_L$  does not cover  $\lambda$ ;
- ▶ Low prefers  $I_L$  and effort 1 with probability 1 to the lottery;



# Alternative approaches to inattention

**Cost of learning about the state** (Wei, 2021):

- ▶ Costly entropy reduction about the state;
- ▶ Flexible information acquisition about  $\theta$ ;

*Interpretation:* **Delegated learning** (e.g., analyzing a dataset).

**Cost of processing the message** (Bloedel and Segal, 2021):

- ▶ Costly entropy reduction about the message;
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**Cost of exposure:**

- ▶ Cost  $\lambda k(e)$  is independent of the informativeness;
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Thanks!

# Conclusion

- ▶ With inattention and private information:
  1. Persuasion mechanisms are equivalent to signals;
  2. Upper censorships are optimal;
  3. Persuasion mechanisms are without loss under deterministic coordination mechanisms;
  4. Stochastic coordination mechanisms can correlate the signal with the attention recommendation.
- ▶ In the paper:
  1. Comparative statics wrt  $\lambda$ ;
  2. Persuasion mechanisms are without loss if  $c$  is known and  $k$  is isoelastic;
  3. “Experimental mechanisms”: a subset of direct and IC stochastic coordination mechanisms that is without loss.

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Thanks!

**Extra slides**

# Setting

Let  $X := [0, 1]$  be the state space, and  $F_0$  a continuous prior CDF on  $X$  with mean  $x_0 \in (0, 1)$ .

## Definition 1 (Information Policies)

Let  $\mathcal{C}(X)$  be the space of real-valued continuous functions on  $X$  endowed with the supremum norm  $|\cdot|_\infty$ . The uninformative policy is  $\underline{l}(x) := (x - x_0)_+$ , and the fully informative policy is  $l_0(x) := \int_0^x F_0(t) dt$ . The set of feasible information policies is

$$\mathcal{I} := \{I \in \mathcal{C}(X) \mid I \text{ is convex, } \underline{l} \leq I \leq l_0, \partial I(X) \subseteq [0, 1]\}.$$

For any  $I \in \mathcal{I}$ , let  $I'$  denote its right-derivative.

## Lemma 3 (Lemma 1)

The mapping  $T: F \mapsto (x \mapsto \int_0^x F(y) dy)$  is a homeomorphism from Bayes-plausible posterior-mean CDFs under the  $L^1$  norm to  $\langle \mathcal{I}, |\cdot|_\infty \rangle$ .

# Equilibrium

Let  $T := X \times \mathbb{R}_+$  be the type space endowed with a probability measure  $P$ . Given  $I \in \mathcal{I}$  and effort  $e \in [0, 1]$ , define the effective posterior-mean CDF  $(e \odot I') := eI' + (1 - e)\bar{F}$ , where  $\bar{F}(x) := \mathbf{1}\{x \geq x_0\}$ .

## Definition 2 (Equilibrium)

An equilibrium is a tuple  $\langle I^*, e, \alpha \rangle$ , where  $I^* \in \mathcal{I}$ ,  $e: T \times \mathcal{I} \rightarrow [0, 1]$  is measurable, and  $\alpha: T \times X \rightarrow [0, 1]$  is measurable, such that:

1. **Action Optimality:** For all  $x \in X$  and  $(c, \lambda) \in T$ ,

$$\alpha(c, \lambda, x) \in \operatorname{argmax}_{b \in [0, 1]} b(x - c).$$

2. **Effort Optimality:** For all  $(c, \lambda) \in T$  and  $\hat{I} \in \mathcal{I}$ ,

$$e(c, \lambda, \hat{I}) \in \operatorname{argmax}_{\tilde{e} \in [0, 1]} \int_X (x - c)_+ d(\tilde{e} \odot \hat{I}')(x) - \lambda k(\tilde{e}).$$

3. **Sender Rationality:**  $I^* \in \operatorname{argmax}_{\hat{I} \in \mathcal{I}} \hat{W}(\hat{I}, e, \alpha)$ , where

$$\hat{W}(\hat{I}, e, \alpha) := \int_T \int_X \alpha(c, \lambda, x) d(e(c, \lambda, \hat{I}) \odot \hat{I}')(x) dP(c, \lambda).$$

# Receiver's Value and the Extensive Margin

## Definition 3

The net informativeness operator is  $\Delta: \mathcal{I} \rightarrow \mathcal{C}(X)$ , defined by  $\Delta I := I - \underline{I}$ . The receiver's value function  $V_\lambda: [0, 1] \times \mathbb{R}_+ \rightarrow \mathbb{R}$  is:

$$V_\lambda(e, \Delta I(c)) := e\Delta I(c) - \lambda k(e).$$

Let  $E_\lambda(\Delta I(c)) := \operatorname{argmax}_{e \in [0, 1]} V_\lambda(e, \Delta I(c))$ .

## Lemma 4 (Lemmata 2 and 3)

The marginal benefit of effort  $B(I, c) := \int_X (x - c)_+ dI'(x) - (x_0 - c)_+$  satisfies  $B(I, c) = \Delta I(c)$ . Furthermore:

1.  $V_\lambda(e, \Delta I(c))$  exhibits strictly increasing differences in  $(e, \Delta I(c))$ .
2. Every selection from the correspondence  $E_\lambda(\cdot)$  is nondecreasing.
3. Given an equilibrium  $\langle I^*, e(\cdot), \alpha \rangle$  and  $\lambda \geq 0$ , the active set  $\{c \in X \mid e(c, \lambda, I^*) > 0\}$  is an interval, and contains  $x_0$  if non-empty.

# Optimality of Upper Censorships

## Assumption 1 (SPness)

For all  $\lambda \geq 0$ , the conditional distribution of  $c$  given  $\lambda$  admits an absolutely continuous density  $g(\cdot \mid \lambda)$ . There exists a peak  $p \in (0, 1)$  such that, for all  $\lambda$ ,  $g(\cdot \mid \lambda)$  is nondecreasing on  $[0, p]$  and nonincreasing on  $[p, 1]$ .

## Definition 4

For any state  $\bar{\theta} \in X$ , the  $\bar{\theta}$ -upper censorship  $I_{\bar{\theta}} \in \mathcal{I}$  is the unique information policy satisfying:

$$I_{\bar{\theta}}(x) = \begin{cases} I_0(x), & \text{if } x \in [0, \bar{\theta}] \\ \max\{I_0(\bar{\theta}) + F_0(\bar{\theta})(x - \bar{\theta}), \underline{I}(x)\}, & \text{if } x \in (\bar{\theta}, 1]. \end{cases}$$

## Proposition 6 (Proposition 2)

Let Assumption 1 hold. There exists an equilibrium  $\langle I^*, e(\cdot), \alpha \rangle$  wherein  $I^*$  is a  $\bar{\theta}$ -upper censorship. Moreover, for any two equilibria  $\langle I_1^*, e_1, \alpha_1 \rangle$  and  $\langle I_2^*, e_2, \alpha_2 \rangle$ ,  $\hat{W}(I_1^*, e_1, \alpha_1) = \hat{W}(I_2^*, e_2, \alpha_2)$ .

# Application

**Setup:** Sender's utility is  $U_S(a, e) = a + \gamma e$  for  $\gamma \geq 0$ . Cost is linear:  $k(e) = e$ . Let  $V_S(I, e, \alpha)$  denote the Sender's ex-ante expected utility under  $U_S$ .

## Definition 5

An information policy  $I \in \mathcal{I}$  is a bi-upper censorship if there exist  $0 \leq \theta_1 \leq x_1 \leq x_2 \leq 1$  and  $m = \frac{\underline{I}(x_2) - [l_0(\theta_1) + F_0(\theta_1)(x_1 - \theta_1)]}{x_2 - x_1}$  such that:

$$I(x) = \begin{cases} l_0(x), & \text{if } x \in [0, \theta_1] \\ l_0(\theta_1) + F_0(\theta_1)(x - \theta_1), & \text{if } x \in (\theta_1, x_1] \\ \underline{I}(x_2) - m(x_2 - x), & \text{if } x \in (x_1, x_2] \\ \underline{I}(x), & \text{if } x \in (x_2, 1]. \end{cases}$$

## Proposition 7 (Proposition 3)

Let  $P(X \times \{\lambda^*\}) = 1$  for some  $\lambda^* \geq 0$ . Let Assumption 1 hold with  $p \geq x_0$ . For every equilibrium  $\langle I^*, e(\cdot), \alpha \rangle$ , there exists a bi-upper censorship  $\hat{I} \in \mathcal{I}$  such that  $V_S(\hat{I}, e, \alpha) \geq V_S(I^*, e, \alpha)$ .

# Coordination mechanisms

## Definition 6

A coordination mechanism is a tuple  $G = (R, \mathcal{R}, M, \mathcal{M}, \pi)$  where  $(R, \mathcal{R})$  is a measurable space of reports,  $(M, \mathcal{M})$  is a measurable space of messages, and  $\pi: R \rightarrow \Delta(\mathcal{I} \times M)$  is a Markov kernel.  $G$  is *direct* if  $(R, \mathcal{R}) = (T, \mathcal{B}_T)$  and  $(M, \mathcal{M}) = ([0, 1], \mathcal{B}_{[0,1]})$ .

## Definition 7

A participation strategy  $(\rho, \delta)$  consists of measurable functions  $\rho: T \rightarrow R$  and  $\delta: T \times M \rightarrow [0, 1]$ . It is an equilibrium in  $G$  if no type  $t \in T$  prefers any measurable deviation  $\eta: M \rightarrow [0, 1]$  and a report  $r \in R$ . A direct mechanism is Incentive Compatible (IC) if truth-telling and obedience  $(\rho^*(t) = t, \delta^*(t, m) = m)$  forms an equilibrium.

## Proposition 8 (Proposition 4)

For every coordination mechanism  $G$  and every equilibrium  $(\rho, \delta)$  in  $G$ , there exists an IC direct coordination mechanism  $F = (T, [0, 1], \pi^F)$  such that  $V_0^F(\rho^*, \delta^*) = V_0^G(\rho, \delta)$ , in which  $V_0$  denotes the Sender's ex-ante expected utility.

# Deterministic mechanisms

A *product coordination mechanism* is a coordination mechanism  $(R, M, \pi)$  such that, for all  $r \in R$ , the probability measure  $\pi(\cdot|r)$  is a product measure.

## Proposition 9 (Proposition 5)

For every product coordination mechanism  $G$ , for every equilibrium  $(\rho, \delta)$  in  $G$ , there exists an IC persuasion mechanism  $I_\bullet$  and a payoff  $v \in \mathbb{R}_+$  such that:  $I_\bullet$  induces  $v$  and  $v \geq V_0^G(\rho, \delta)$ .

## Experimental and activation mechanisms

A direct coordination mechanism  $G = (T, E, \pi)$  is an *experimental mechanism* if: for all  $t$ , there exist a probability measure  $\mu_t$  on  $\mathcal{I}$  and a measurable function  $f_t: \mathcal{I} \rightarrow E$  such that, for all  $\bar{I} \in \mathcal{B}_{\mathcal{I}}$ ,  $\bar{E} \in \mathcal{B}_E$ , we have

$$\pi(\bar{I} \times \bar{E} | t) = \mu_t(f_t^{-1}(\bar{E}) \cap \bar{I}).$$

### Proposition 10 (Experimental mechanisms are without loss)

For every direct incentive compatible coordination mechanism  $G$ , there exists an incentive compatible experimental mechanism  $F$  such that

$$V_0^F(\rho^*, \delta^*) = V_0^G(\rho^*, \delta^*).$$

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A direct coordination mechanism  $G = (T, E, \pi)$  is an *activation mechanism* if: for all  $t$ , there exist a probability measure  $\mu_t$  on  $E$  and a measurable function  $f_t: E \rightarrow \mathcal{I}$  such that, for all  $\bar{I} \in \mathcal{B}_{\mathcal{I}}$ ,  $\bar{E} \in \mathcal{B}_E$ , we have

$$\pi(\bar{I} \times \bar{E} | t) = \mu_t(f_t^{-1}(\bar{I}) \cap \bar{E}).$$

### Proposition 11

For every direct incentive compatible coordination mechanism  $G$ , there exists an incentive compatible activation mechanism  $F$  such that

$$V_0^F(\rho^*, \delta^*) = V_0^G(\rho^*, \delta^*)$$

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